Tablet Compression Tooling Inspection

A structured worksheet for recording observed punch and die conditions, gathering evidence, and routing quality-impact decisions to the site's qualified technical and quality teams.

Template · not an approved SOP | Version 1.1 · 9 November 2026 | Author: James Demetro

This is a general documentation aid, not a validated defect classifier, product-specific risk assessment, engineering disposition, or regulatory determination. It does not define universal tooling limits. Follow the current site SOP, approved tooling drawings, product controls, and QA procedures. A qualified site team must review evidence and authorize any disposition.

What is a tablet compression tooling inspection worksheet?

A tablet compression tooling inspection worksheet is a record for documenting observed punch and die conditions, measurements, supporting evidence, and site review. Use it to organize an investigation; it does not decide whether tooling is acceptable or safe to return to production.

Use this framework to organize observations when tooling shows sticking, picking, wear, surface damage, corrosion, deformation, or suspected dimensional change. Connect each symptom to tool identity, press position, process context, inspection evidence, and documented follow-up.

It does **not** determine whether a punch or die is acceptable for use. Tool acceptance criteria, measurement methods, repair limits, coating selection, and return-to-service authorization must come from applicable drawings, supplier specifications, equipment instructions, and the site's approved quality system.

Risk scoring is site-defined. ICH Q9(R1) describes quality risk management principles and example methods, including FMEA. It does not prescribe a universal Severity × Probability × Detectability scale or a universal action threshold for tablet tooling. Define and approve any scoring model locally before use.

System and items to identify

- Tablet press, model, turret/station, tooling set ID, and controlled drawing revision
- Configuration: single-tip or multi-tip; tips per punch; site/vendor tip numbering and orientation convention
- For a multi-tip observation, record affected punch (upper/lower), station, individual tip position, orientation, and matching die ID
- Punch and die identifiers, serial/lot numbers, and drawing revision
- Tool steel, surface treatment or coating, if documented
- Product/formulation and batch context, handled under site confidentiality controls
- Observed defect, date/time, operator report, and inspection method

Capture evidence before disposition

- Photographs with scale and orientation where practical
- Measured values, units, instrument ID, calibration status, and method
- Tool history, cleaning/maintenance history, and prior repairs
- Relevant process records and tablet defect observations
- Reviewer, date, decision rationale, actions, and closure evidence

Tooling conditions to investigate

These are prompts for evidence collection, not diagnostic conclusions. Similar tablet defects can have tooling, formulation, process, environmental, or combined causes.

Observed symptom	Tooling-related possibilities to examine	Evidence / questions
Sticking or filming Material appears adhered to a punch face.	Surface finish or damage, residue, wear, coating condition, punch-face geometry, or an interaction with formulation and process conditions.	Where on the face? Does it repeat by station? What do cleaning records, surface inspection, roughness measurement, and process trends show?
Picking Material is retained in engraved or recessed regions.	Logo/engraving geometry, local surface condition, buildup, coating integrity, punch condition, or formulation/process interactions.	Map the retained material to the engraving. Record image, affected stations, repeatability, and inspection of edges, radii, and surface.
Capping or lamination Tablets separate into layers or cap.	Tool wear or damage may be one factor among several; consider tooling geometry and condition alongside formulation, air entrapment, dwell time, and compression settings.	Compare defect timing and station pattern with punch tips, tip lands, die condition, process records, and product/test data.
Wear, corrosion, or deformation	Material compatibility, cleaning/storage conditions, use history, mechanical loading, maintenance, or repair history.	Record location and extent; compare to the controlled drawing/specification using an approved measurement method.

Open a structured evidence checklist

- Identify if the observation is isolated to one station/tool or appears across multiple stations.
- Record whether the issue is new, recurring, increasing, or associated with a process/material change.
- Use approved cleaning and inspection methods; preserve samples or images according to site procedures.
- Separate measured facts from hypotheses. Label each hypothesis and list what evidence would confirm or rule it out.
- Route potential product-quality impact through the site's deviation, investigation, and change-control processes.

Record the condition against controlled requirements

Use the approved tool drawing/specification and site method to define the feature, measurement location, units, and limit. "Pass/fail" below is a recorder field only; this worksheet supplies no acceptance criteria. Select one result after applying your controlled criteria.

Inspect	Feature / location detail	Measured value / units	Drawing/spec limit & revision	Instrument ID	Calibration due date	Pass / fail / N/A	Inspector initials
[ ]	**Punch tip face** _____	_____	_____	_____	__/__/____	Pass / Fail / N/A / Pending review	_____
[ ]	**Tip land** _____	_____	_____	_____	__/__/____	Pass / Fail / N/A / Pending review	_____
[ ]	**Barrel** _____	_____	_____	_____	__/__/____	Pass / Fail / N/A / Pending review	_____
[ ]	**Head** _____	_____	_____	_____	__/__/____	Pass / Fail / N/A / Pending review	_____
[ ]	**Die bore** _____	_____	_____	_____	__/__/____	Pass / Fail / N/A / Pending review	_____
[ ]	**Die table wear** _____	_____	_____	_____	__/__/____	Pass / Fail / N/A / Pending review	_____
[ ]	**Engraving / logo condition** _____	_____	_____	_____	__/__/____	Pass / Fail / N/A / Pending review	_____

The checkbox means the item was inspected; it does not mean it passed. Record the method, measurement location, units, drawing/specification identifier and revision, instrument calibration status, and any exception in the controlled record. This browser worksheet does not save or transmit entries.

Choose a method that matches the feature and controlled specification

These are common instrument examples, not a mandated method. Verify suitability, resolution, setup, cleanliness, calibration status, and measurement uncertainty for the actual tolerance and site procedure. Drawing notes and the tool supplier's method take precedence.

Dimension / feature	Possible instrument	Typical measurement approach	Record with result
Tip diameter	Calibrated outside micrometer; optical comparator for profiled tips	Measure at the drawing-defined axial location and orientations; avoid burrs, deposits, or damaged edges. Capture orientation and repeat readings per site method.	Location, orientation, units, instrument ID, drawing revision.
Overall punch length	Length comparator, height gauge with suitable fixture, or calibrated length-measuring system	Seat the punch against specified datum surfaces and measure the drawing-defined overall length without rocking or loading the tip.	Datum/setup, fixture, units, instrument ID.
Working length	Tooling-specific working-length gauge or calibrated length comparator/measurement system specified by the drawing	Measure from the drawing-defined head-flat datum to the specified point on the cup/face. Terminology and method may vary by tooling standard and drawing; confirm before comparing. Working length is associated with punch penetration and tablet thickness/consistency; tablet weight is also governed by fill and process factors.	Tool standard, datum, tip/cup geometry, method, units, instrument ID.
Punch tip-to-die clearance / interference check	Approved tooling inspection fixture or specified dimensional measurement system	Use only the controlled supplier/site procedure for the matched punch and die. Do not force a punch into a die as a generic fit test. Stop if resistance, scoring, burrs, or damage is observed and escalate for qualified review.	Punch/die IDs, orientation, observed condition or specified measured result, method.
Barrel diameter	Outside micrometer	Measure at specified axial bands and orientations; separately document any taper, out-of-round, scoring, or local wear required by the drawing.	Axial location and angular orientation for each reading.
Tip land width	Optical comparator / vision measurement system; suitable microscope with calibrated scale	View the specified tip profile, identify the drawing-defined land boundaries, and measure at designated positions. Do not infer boundaries from appearance alone.	Profile view, magnification/calibration, measurement locations.
Die bore	Two-point or three-point bore gauge; air gauge where specified; calibrated plug gauges for go/no-go checks only when authorized	Set/verify the gauge using a traceable master or reference; measure drawing-defined locations and directions. Use go/no-go limits only from the applicable controlled specification. Never force a plug gauge into a tight bore; stop if it binds and document the observation. Record discrete values when required.	Gauge setting reference, depth, orientation, units, instrument ID.
Concentricity / runout relationship	Dial test indicator on a suitable datum fixture; optical/CMM system if specified	Establish the drawing datum, support and rotate the part using the defined setup, then record total indicator reading or specified geometric result. Do not substitute runout for a different GD&T characteristic without approval.	Datum, fixture, rotation/setup, indicated characteristic and units.
Surface roughness (when specified)	Contact surface profilometer or validated optical profilometer suitable for the material/feature	Use the controlled trace direction, cutoff/filter/evaluation length and sampling location from the drawing or method. Small punch-face geometry may require an appropriate stylus or optical method.	Parameter (e.g., Ra/Rz), trace direction, filter/cutoff, locations.

Plug-gauge caution: Use only a gauge identified and authorized for the bore and its controlled limits. A go/no-go result is meaningful only for the defined gauge and inspection method. Do not force a gauge; stop if it binds and refer the condition to the qualified site reviewer.

Measurement caution: A numeric reading without its datum, location, method, units, instrument status, and applicable controlled limit may not be comparable or decision-ready. Route uncertain or out-of-limit results through site procedures.

Sticking reported at station 12

Illustrative example only — not acceptance criteria. All values and batch details below are invented solely to demonstrate how to document evidence. They are not a real event, measurement, recommendation, or disposition.

Record field	Illustrative entry
Observed pattern	Sticking reported on tablets associated with station 12 during a routine run; adjacent stations are recorded as a comparison group. Operator reports visible residue near a small region of the upper-punch face. This observation does not establish cause.
Batch context	Example batch EX-001; product identifier omitted; observation noted during the middle third of the run after a documented pause. Compare approved batch/process records and product impact under site procedures.
Example measurement	Punch face: Ra 0.38 µm at an example trace location, measured with fictional profilometer PROF-EX-014 (calibration due 28 Feb 2027). No drawing/specification limit is supplied here; result cannot be called pass or fail from this worksheet.
Evidence to collect	Photograph with orientation and scale; trace file and settings; station/tool serials; neighboring-station comparison; cleaning/maintenance history; retained residue sample if site procedure allows; process and batch records; approved drawing and supplier specification.
Sample FMEA-style score	**S=4 × P=3 × D=2 → illustrative product 24.** Example rationale: S=4 because the team provisionally considers potential product-quality impact significant; P=3 because recurrence is not yet established; D=2 because the example controls are assumed likely to detect the symptom. These are made-up ratings and rationale for demonstrating documentation only.
Example follow-up actions	Open/route the event under site procedure; QA/Engineering assess affected product and containment; verify the reported station/tool identity; inspect and measure against controlled criteria; compare affected and unaffected stations; examine formulation/process alternatives; document the approved disposition and post-action verification.

Scoring scales, definitions, weighting, decision thresholds, and escalation rules are site-defined. The example number 24 has no meaning outside a locally approved method and is not a tool or product acceptance decision.

Published examples for orientation only

These are supplier-published examples for named material conditions or coating systems, not universal material properties and not a recommendation for pharmaceutical tooling. They do not establish suitability for product-contact use, compatibility, cleanability, regulatory status, or acceptance criteria.

Material / treatment example	Published property example	Thickness / case example	Source and limitation
D2 tool steel	One heat-treatment schedule reports 710–760 HV (60–62 HRC); other schedules in the same datasheet differ.	Not applicable to base steel.	Taylor Special Steels D2 datasheet. Values vary by heat treatment and condition. Values vary by supplier and process; verify against current supplier datasheet. Not acceptance criteria.
S7 tool steel	One supplier reports typical heat-treated hardness around 55–58 HRC, depending on tempering condition.	Not applicable to base steel.	Interlloy S7 datasheet. Datasheet notes approximate values dependent on heat treatment and quenching. Values vary by supplier and process; verify against current supplier datasheet. Not acceptance criteria.
Cemented tungsten carbide	One named supplier’s carbide grade family lists 580–1,350 HV30 across different grades/compositions; this is not a generic carbide range.	Not applicable to base material.	Hyperion grade data. Supplier labels these typical values; application shown is carbide-roll grades. Values vary by supplier and process; verify against current supplier datasheet. Not acceptance criteria.

Material / treatment example	Published property example	Thickness / case example	Source and limitation
Hard chromium plating	One supplier's general industrial hard-chrome reference reports 850–1,050 HV.	That supplier describes 25–250 µm as a typical wear-deposit range; other thicknesses are used for different applications.	NORAMARK hard chrome process reference. General industrial plating figures; not a tablet-punch specification. Values vary by supplier and process; verify against current supplier datasheet. Not acceptance criteria.
TiN PVD coating	Oerlikon BALINIT A product: microhardness 30 ± 3 GPa.	Typical food-processing coating thickness listed as 1–4 µm for that product.	Oerlikon BALINIT food-processing coating sheet. Manufacturer says values are approximate and depend on application, environment, and test conditions. Values vary by supplier and process; verify against current supplier datasheet. Not acceptance criteria.
CrN PVD coating	Oerlikon BALINIT CNI product: microhardness 18 ± 3 GPa.	Typical food-processing coating thickness listed as 2–8 µm for that product.	Oerlikon BALINIT food-processing coating sheet. Manufacturer says values are approximate and depend on application, environment, and test conditions. Values vary by supplier and process; verify against current supplier datasheet. Not acceptance criteria.
Nitriding / gas nitrocarburizing reference	For C45E (1.1191), one processor lists surface hardness 300–500 HV0.5 for its nitriding process.	The same technical sheet lists maximum nitriding hardness depth of 0.7 mm for that material/process.	Lingenhöle nitriding material data. Material- and process-specific C45E example; not a generic D2/S7 or punch-tooling value. Values vary by supplier and process; verify against current supplier datasheet. Not acceptance criteria.

Do not use this table to select a coating, set a drawing limit, or approve a tool. Confirm exact grade, substrate, heat treatment, coating process/product, dimensional allowance, test method, and intended use with the current supplier documentation and qualified site functions.

Terms used in tooling observations

Sticking
Tablet material adheres to a punch face or other tooling surface during compression/ejection.

Picking
Localized sticking in engraved, embossed, or recessed punch features, which may pull material from the tablet surface.

Capping
Separation of a tablet's crown or cap from the body, often observed during ejection or handling; causes can be multifactorial.

Lamination
Separation of a tablet into two or more layers or horizontal planes.

Filming
A thin residue layer or film accumulating on a tooling surface; teams may use the term differently, so document the observed condition.

Tip land
The narrow peripheral region at the punch tip surrounding the cup/profile, as defined by the tool drawing.

Barrel
The punch body section guided within the press turret/bushing, with boundaries and features defined by tooling design.

Head
The punch end engaged by press cams/rollers; terminology and geometry can vary by punch design.

Cup depth

Depth of the concave punch-tip cup measured from a defined reference plane to a specified point/feature under the applicable drawing method.

Land wear

Wear, rounding, deformation, or dimensional change observed at the tip land; assessment requires comparison with controlled design requirements.

Chrome flaking

Visible separation or loss of a chromium-containing coating/deposit. Record location, extent, images, and substrate exposure; do not infer product impact or disposition from appearance alone.

Text layout suitable for an infographic

Use this as an investigation-routing aid, not as an automated diagnostic tree or production instruction.

```
OBSERVED TABLET DEFECT
        |
        +-- Sticking / filming
        |       Inspect: punch tip face, finish/damage, residue, station pattern;
        |                review cleaning history and approved process/formulation records.
        |       Record: exact station/tool IDs, photo orientation, trace settings,
        |                repeatability, affected samples/batch context.
        |
        +-- Picking / logo retention
        |       Inspect: engraving/logo edges and recesses, local face condition,
        |                coating condition if documented, and comparable stations.
        |       Record: logo-area images, affected characters/features, tool drawing,
        |                cleaning evidence, and measured results using controlled method.
        |
        +-- Capping / lamination
        |       Inspect: punch-tip profile, tip land, head/tip damage, die condition;
        |                review tooling alongside non-tooling hypotheses.
        |       Record: defect timing, station distribution, tablet observations,
        |                process and material context available under site procedure.
        |
        +-- Dimensional / visual wear, corrosion, deformation, coating loss
                Inspect: affected feature, adjacent surfaces, matched upper/lower tools,
                         die bore/table interface, and prior repair/coating records.
                Record: scale photographs, location/extent, drawing revision, measurement
                         method/instrument/calibration, and controlled comparison.

FOR EVERY BRANCH:
Separate observed fact from hypothesis → preserve/identify evidence → assess potential
product impact and containment under site procedures → escalate to Engineering and QA
for investigation, risk assessment, and authorized disposition → document verification.
No branch in this worksheet authorizes continued use, repair, coating, or release.
```

Document the site’s assessment

Complete one row per failure mode or condition using the site-approved risk procedure. Attach the controlled scale and acceptance criteria used. Do not treat blank scores as zero risk.

Failure mode / effect	Evidence and cause hypothesis	Current controls	Site-approved risk score / rationale	Owner, action, due date
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____

Suggested methods and terminology must be selected by the site's quality system. Any scoring, ranking, or residual-risk conclusion should record its rationale, assumptions, reviewers, and approval.

Use controlled criteria for repair, coating, or retirement

Inspection or measurement

Compare measured condition with the current approved drawing/specification and the tool supplier's instructions. Record instrument, method, units, calibration status, and uncertainty where relevant.

Polishing or restoration

Confirm authorized repair limits and the post-repair inspection plan with the responsible engineering and quality functions. Do not infer that polishing is acceptable from a visual defect alone.

Coating or surface modification

Review the exact coating supplier, process, substrate compatibility, dimensional impact, cleanliness, documentation, and intended product-contact use. Use the current supplier technical data and site qualification/change-control process.

Retirement or replacement

Use controlled acceptance/rejection criteria and document tool identity, reason, impact assessment, and replacement/verification record under site procedures.

No universal coating-property values are shown here. Hardness, friction, temperature capability, thickness, and release behavior vary by coating formulation, deposition process, test method, substrate, and service conditions. Obtain values from the supplier's applicable datasheet and evaluate them for the intended application.

Keep observations, analysis, and decisions traceable

1. **Describe the observation.** Record the symptom, affected station/tool, date, batch context, and supporting images or measurements.
2. **Establish the evidence.** Check the relevant controlled drawing, tool history, inspection records, cleaning/maintenance records, and process information.
3. **Assess quality risk.** Use the site-approved method and involve the functions required by local procedures. Note uncertainties and alternative causes.
4. **Approve a controlled action.** Engineering and Quality determine inspection, repair, coating, replacement, containment, or further investigation under the site quality system.
5. **Verify and close.** Record post-action results, return-to-service authorization, change/deviation references, and evidence that actions were completed.

Ownership, review, and revision history

Suggested title / slug: Tablet Compression Tooling Inspection & Risk Review Worksheet — `tablet-compression-tooling-inspection-risk-review-worksheet`

Author: James Demetro

Credentials (if applicable and verified):

_____

Last-reviewed date:
__/__/____

Site / department:

_____

Document owner:

_____

Quality reviewer:

_____

Local controlled document reference (if assigned):

_____

Review status: Template draft; not yet reviewed or approved by a named qualified site reviewer. James Demetro is listed as author only; no credentials or approval are implied.

Version	Date	Change summary	Author / reviewer	Approval / status
0.1	9 Nov 2026	Initial discussion worksheet draft.	Not assigned	Uncontrolled draft; not an approved SOP.
0.2	9 Nov 2026	Added inspection checklist, measurement guide, illustrative example, supplier-reference table, glossary, investigation flowchart, and document-control fields. Preserved caution and no-universal-limit framing.	Unassigned	Draft for qualified site review; no approval implied.
1.0	9 Nov 2026	Consolidated the worksheet as an adaptable template, clarified supplier-source limitations, added named authorship, and retained pending-review status.	James Demetro	Template draft — not reviewed or approved.
1.1	9 Nov 2026	Added multi-tip tool identification prompts, working-length and punch-to-die inspection records, plug-gauge cautions, clearer answer-first summary, metadata, and descriptive JSON-LD. No universal acceptance limits added.	James Demetro	Template draft — not reviewed or approved.

Do not label this worksheet an approved SOP until reviewed, approved, and controlled by the appropriate organization. Add credentials or a review date only after they are verified and documented.

Regulatory context and technical property sources

- FDA, Process Validation: General Principles and Practices (2011). Nonbinding guidance describing lifecycle process validation principles.
- ICH Q9(R1), Quality Risk Management (Step 4, 2023). General quality risk-management principles and example tools.
- FDA / ICH Q13, Continuous Manufacturing of Drug Substances and Drug Products (2023). Relevant only where continuous manufacturing applies.
- Taylor Special Steels, D2 cold-work tool steel datasheet. Example heat-treatment hardness values.
- Interlloy, S7 tool steel datasheet. Approximate hardness depends on treatment and quenching.
- Hyperion Materials & Technologies, cemented carbide grade data. Named grade examples for a supplier-specific application.
- NORAMARK, hard chrome plating process reference. General industrial plating values.
- Oerlikon BALINIT coatings for food processing and packaging. Product-specific coating examples; manufacturer notes data are approximate.
- Lingenhöle, nitriding material data. C45E-specific process example.

Regulatory references provide broad quality-system context. The supplier references support the narrowly described illustrative property examples only. None establishes punch/die limits, surface-roughness thresholds, coating specifications, suitability for a particular pharmaceutical product, or defect-specific acceptance criteria. Always use current controlled supplier documentation and site-approved criteria.